

Computer Graphics (CSIT 401)

IV semester, BScCSIT

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Syllabus

Unit	Contents	Hours	Remarks
1.	Introduction to Computer Graphics	5	
2.	Scan Conversion Algorithms	7	<i>Very Important</i>
3.	2D Transformations and Viewing	6	<i>Very Important</i>
4.	3D Transformations and Viewing	5	
5.	3D Object Representation	6	
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Unit 8

Illumination, Shading, and Rendering Techniques

(5 hrs.)

Introduction to light and color models: RGB, CMY, HSV, illumination models: Ambient, diffuse, specular, Phong reflection, Polygon rendering techniques: Flat shading, Gouraud shading, Phong shading, Fast Phong Shading.

Unit 8

Section 1: Illumination Models, Light Sources, Basic Illumination Models: Ambient light

Illumination and surface rendering model

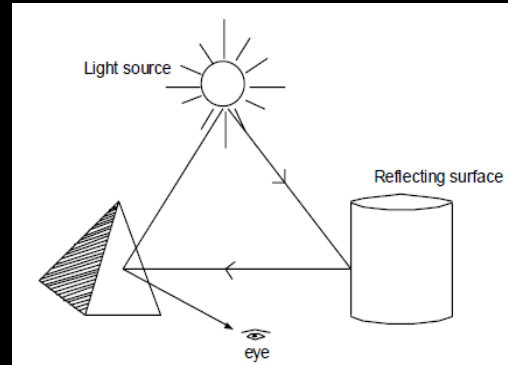
- Once visible surface has been identified by hidden surface algorithm, a shading model is used to compute the intensities and color to display for the surface. For realistic displaying of 3d scene it is necessary to calculate appropriate color or intensity for that scene.
- **Illumination model or a lighting model** is the model for calculating light intensity at a single surface point. Sometimes also referred to as a **shading model**.
- An illumination model is also called **lighting model** and sometimes called as a **shading model** which is used to calculate the intensity of light that we should see at a given point on the surface of a object.
- A surface-rendering algorithm uses the intensity calculations from an illumination model.

Components of Illumination model

- **Light Sources:** type , color, and direction of the light source
- **Surface Properties :** reflection, opaque/transparent, shiny/dull.

Light Source

- Object that radiates energy are called light sources, such as sun, lamp, bulb, fluorescent tube etc.
- Sometimes light sources are referred as light emitting object and light reflectors. Generally light source is used to mean an object that is emitting radiant energy e.g. Sun.
- ***Total Reflected Light = Contribution from light sources + Contribution from reflecting surfaces***



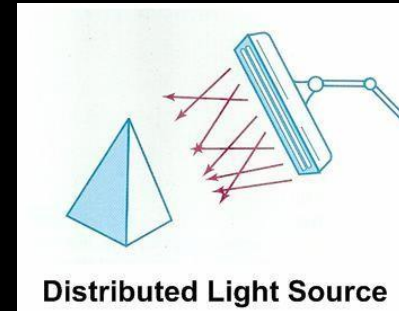
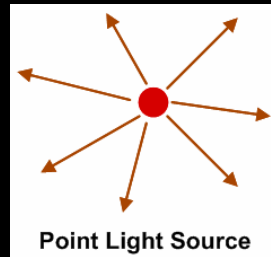
Light Source

Point Source:

Point source is the simplest light emitter e.g., light bulb. When light source model is a reasonable approximation for sources whose dimensions are small compared to the size of objects in a scene.

Distributed Light Source:

area of source is not small compared to the surfaces in the scene e.g., Fluorescent lamp



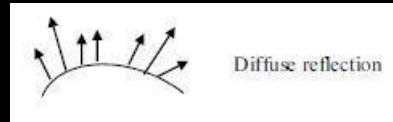
Light Source

- When light is incident on opaque surface part of it is reflected and part of it is absorbed.

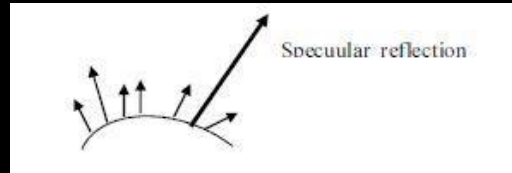
$$\therefore I = A + R$$

- The amount of incident light reflected by a surface depends on the type of material. Shining material reflects more incident light and dull surface absorbs more of the incident light.
- For transparent surfaces, some of the incident light will be reflected and some will be transmitted through the material.

- When light is incident on an opaque surface part of it is reflected and part of it is absorbed.
- Surface that are rough or grainy, tend to scatter the reflected light in all direction which is called **diffuse reflection**.



- When light sources create highlights, or bright spots, called **specular reflection**



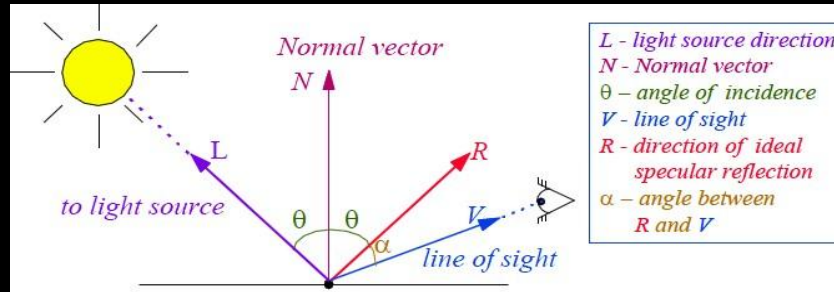
Illumination Models

Illumination models:

Illumination models are used to calculate light intensities that we should see at a given point on the surface of an object. Lighting calculations are based on the optical properties of surfaces, the background lighting conditions and the light source specifications. All light sources are considered to be point sources, specified with a co-ordinate position and an intensity value (color). Some illumination models are:

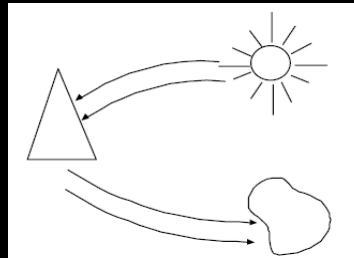
1. Ambient Light
2. Diffuse Reflection
3. Specular Reflection and phong model

NOTE:



1. Ambient light:

- This is a simplest illumination model
- We can think of this model, which has no external light source-self-luminous objects. A surface that is not exposed directly to light source still will be visible if nearby objects are illuminated.
- The combination of light reflections from various surfaces to produce a uniform illumination is called **ambient light or background light**.
- Also called background light
- **Ambient light** means the **light** that is already present in a scene, before any additional **lighting** is added. It usually refers to natural **light**, either outdoors or coming through windows etc. It can also mean artificial **lights** such as normal room **lights**.



1. Ambient light:

- Multiple reflection of nearby (light-reflecting) objects yields a uniform illumination.
- A form of diffuse reflection independent of the viewing direction and the spatial orientation of a surface
- Ambient light has no spatial or directional characteristics and amount on each object is a constant for all surfaces and all directions. In this model, illumination can be expressed by an illumination equation in variables associated with the point on the object being shaded. The equation expressing this simple model is

$$I = K_a$$

Where, I is the resulting intensity and K_a is the **object's intrinsic intensity**.

- If we assume that ambient light impinges equally on all surface from all direction, then

$$I = I_a K_a$$

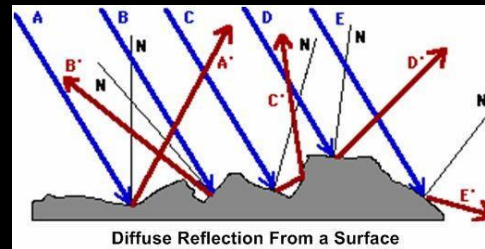
Where I_a is **intensity of ambient light**. The amount of light reflected from an object's surface is determined by K_a , the **ambient-reflection coefficient**. K_a ranges from **0 to 1**.

Unit 8

Section 2: Basic Illumination Models: Diffuse reflection, Specular reflection

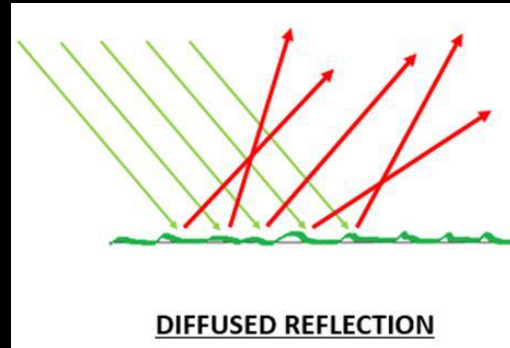
2. Diffuse Reflection

- Objects illuminated by ambient light are uniformly illuminated across their surfaces even though light are more or less bright in direct proportion of ambient intensity.
- Surfaces are rough
- Incident light is scattered with equal intensity in all directions
- Surfaces appear equally bright from all direction.
- Such surfaces are called *ideal diffuse reflectors* (also referred to as *Lambertian reflectors*)
- It is the *type of reflection that occurs when light hits a rough surface and scatters in many directions.*



2. Diffuse Reflection

- Illuminating object by a point light source, whose rays enumerate uniformly in all directions from a single point.
- The object's brightness varies from one part to another, depending on the direction of and distance to the light source.
- Color of an object is determined by the color of the diffuse reflection of the incident light.
- If any object surface is red then there is a diffuse reflection for red component of light and all other components are absorbed by the surface.



2. Diffuse Reflection

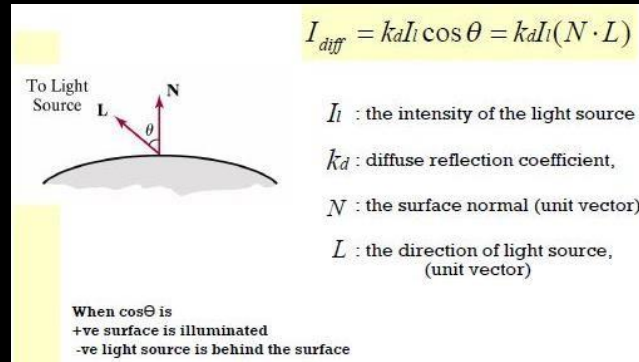
- The *diffuse-reflection coefficient*, or *diffuse reflectivity*, k_d (varying from 0 to 1) define the fractional amount of the incident light that is diffusely reflected.
- The parameter k_d (actually function of surface color) depends on the reflecting properties of material so for highly reflective surfaces, the k_d nearly equal to 1.
- If a surface is exposed only to ambient light, we can express the intensity of the diffuse reflection at any point on the surface as:

$$I_{amb,Diff} = k_d \cdot I_a, \text{ (where } I_a = \text{intensity of ambient light)}$$

2. Diffuse Reflection

- If N is the unit normal vector to a surface and L is the unit direction vector to the point light source from a position on the surface then $\cos \theta = N \cdot L$ (*Lambertian Cosine Law*) and the diffuse reflection equation for single point source illumination is:

$$I_{\text{diff}} = k_d \cdot I_l \cos \theta = k_d I_l (N \cdot L)$$



- We can combine the ambient and point source intensity calculations to obtain an expression for the total diffuse Reflection. Thus, we can write the total diffuse reflection equation as:

$$I_{\text{Diff}} = k_d \cdot I_a + k_d I_l (N \cdot L)$$

Diffuse Reflection: Lambert's Cosine Law:

The intensity of diffuse reflection due to ambient light is;

$$I_{adiff} = k_a I_a$$

The radiant energy from any small surface dA in any direction relative to surface normal is proportional to $\cos \theta$. That is, brightness depends only on the angle θ between the light direction L and the surface normal N .

$\therefore$ Light intensity $\propto \cos \theta$.

If I_l is the intensity of the point light

source and k_d is the diffuse reflection coefficient, then the diffuse reflection for single point source can be written as;

$$I_{pdiff} = k_d I_l \cos \theta$$

$$I_{pdiff} = k_d I_l (N \cdot L)$$

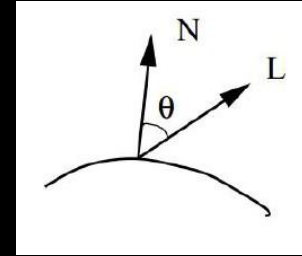


Fig: Angle of incidence θ between the unit light-source direction vector L and the unit surface normal

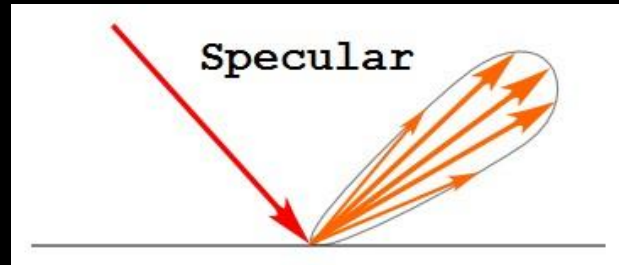
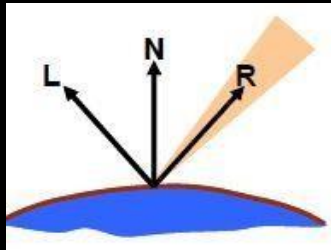
$\therefore$ Total diffuse reflection (I_{diff}) = Diff (due to ambient light) + Diff. due to pt.source.

$$I_{diff} = I_{adiff} + I_{pdiff}$$

$$I_{diff} = k_a I_a + k_d I_l (N \cdot L)$$

3. Specular reflection

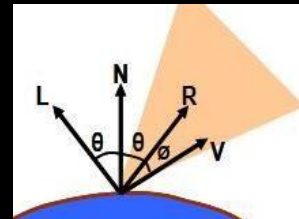
- When we look at an illuminated shiny surface, such as polished metal, a person's forehead, we see a highlight or bright spot, at certain viewing direction. Such phenomenon is called **specular reflection**.
- It is the result of total or near total reflection of the incident light in a concentrated region around the "**specular reflection angle = angle of incidence**".
- Specular reflection, also known as regular reflection, is the mirror like reflection of waves, such as light, from a surface. In this process, **each incident ray is reflected at the same angle to the surface normal as the incident ray**, but on the opposing side of the surface normal in the plane formed by incident and reflected rays. The result is that an image reflected by the surface is reproduced in mirror-like (specular) fashion.



3. Specular reflection

Let SR angle = angle of incidence as in figure.

- N - unit vector normal to surface at incidence point
- R - unit vector in the direction of ideal specular reflection.
- L - unit vector directed to words point light source.
- V - unit vector pointing to the viewer from surface.
- ϕ - the viewing angle relative to the specular reflection direction.



The intensity of specular reflection (I_{spec}) at a surface point is calculated as:

$$I_{\text{spec}} = k_s I_l \cos^{n_s} \phi = k_s I_l (\mathbf{R} \cdot \mathbf{V})^{n_s}$$

where,

I_l : intensity of light source.

n_s : Specular reflection exponent controlling surface shininess.

k_s : Color-independent specular reflection coefficient.

3. Specular reflection and Phong model

- For ideal reflector (perfect mirror), incident light is reflected only in the specular reflection direction.

V & R coincide ($\theta = 0$)

- Shiny surface have narrow θ and dull surface wider θ .
- **The Phong model is a local illumination model used to calculate lighting at a point on a surface.**
- An empirical model for calculating specular-reflection range developed by Phong Bui Tuong-called Phong specular reflection model (or simply **Phong model**), sets the intensity of specular reflection proportional to $\cos^n \theta$: 0 to 90° .
- Specular reflection parameter n_s is determined by type of surface
- Intensity of specular reflection depends upon material properties of the surface and θ . Other factors such as the polarization and color of the incident light.
- For monochromatic specular intensity variations can approximated by SR coefficient $w(\theta)$

3. Specular reflection and Phong model

- **Specular reflection** describes highlights on shiny surfaces.
- Light reflects highly in one dominant direction.
- The **Phong model** provides an empirical formula.
- It approximates this real-world phenomenon mathematically.
- It calculates highlight brightness based on viewer angle.

Alternatively, accounting for varying incident angles:

$$I_{\text{spec}} = w(\theta) I_l \cos^{n_s} \phi$$

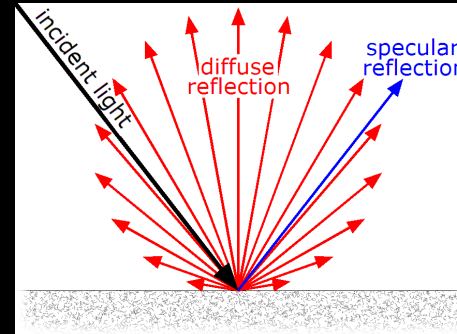
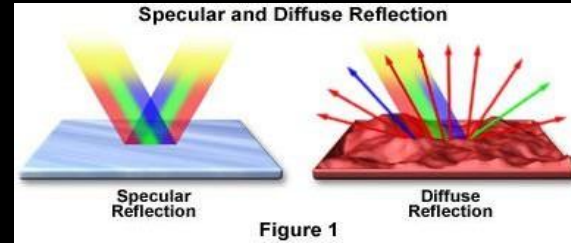
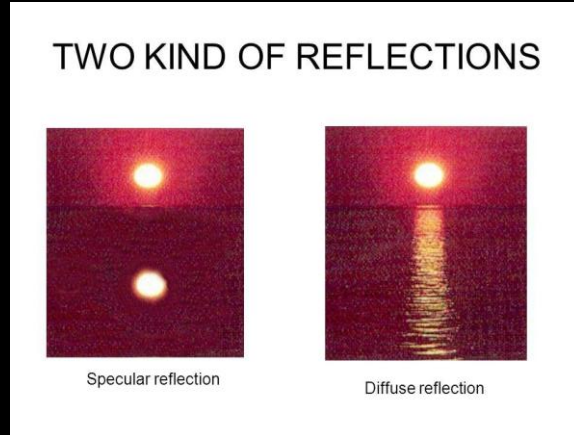
where,

I_l : intensity of light source.

$w(\theta)$: Specular reflection coefficient varying with angle

n_s : Specular reflection exponent controlling surface shininess.

Diffuse Vs Specular Reflection



Diffuse Vs Specular Reflection

Diffuse Reflection	Specular Reflection
Light scatters in all directions.	Light reflects in a concentrated single direction.
The surface texture is rough or matte surfaces (e.g., chalk, cloth, brick).	The surface texture is smooth or glossy surfaces (e.g., polished metal, plastic, mirror).
It is based on Lambert's Cosine Law ($N \cdot L$).	It is based on alignment of reflection vector and view vector ($R \cdot V$).
The key parameters include roughness factor, diffuse color coefficient.	The key parameters include glossiness/shininess exponent (specular power).
The visual effect provides the object's base color and matte shading.	The visual effect creates shiny, bright specular highlights.
Independent of viewer position (looks the same from any angle)	Dependent on viewer position (highlights shift with the camera)

Unit 8

Section 3: Intensity attenuation and Color consideration, Transparency, Shadows

Intensity Attenuation

- Intensity attenuation refers *to the decrease in the brightness or luminance of an object as the distance from the light source increases*. This phenomenon occurs due to the scattering and absorption of light by the medium through which it travels, as well as geometric factors such as the inverse square law.
- As radiant energy from a point light source travels through space, its amplitude is attenuated by the factor $1/d^2$, where d is the distance that the light has traveled. This means that a surface close to the light source (small d) receives higher incident intensity from the source than a distant surface (large d).
- Therefore, *to produce realistic lighting effects, illumination model should take intensity attenuation into account*. Otherwise, we are likely to illuminate all surfaces with same intensity.
- For a point light source *attenuation factor is $1/d^2$* . And for a distributed light source attenuation factor is given by *inverse quadratic attenuation function, $f(d) = 1/(a_0 + a_1d + a_2d^2)$* .

Color Consideration

Color consideration in computer graphics involves accurately representing the colors of objects under various lighting conditions and material properties. This includes factors such as:

1. **Diffuse reflection:** Objects reflect light diffusely, with the color of the object determined by the material's inherent color and the color of the incident light. Diffuse reflection is commonly simulated using the Lambertian reflectance model.
2. **Specular reflection:** Shiny or reflective surfaces exhibit specular reflection, where incoming light is reflected in a concentrated manner, often creating highlights. The color of specular highlights is influenced by both the material's properties and the color of the light source.
3. **Ambient lighting:** Ambient light contributes to the overall illumination of a scene and affects the perceived colors of objects. Ambient light can be simulated using global illumination techniques or by adding an ambient term to lighting calculations.

Transparency

Transparency refers to the *property of allowing light to pass through an object*, enabling objects behind it to be visible to some degree. In computer graphics, transparency is commonly achieved using techniques such as:

- **Alpha blending:** Objects with transparency are rendered by blending their color with the color of the background based on an alpha (*opacity*) *value*. This technique is often used for rendering transparent materials like glass or water.
- **Ray tracing:** Ray tracing algorithms simulate the behavior of light rays as they interact with objects in a scene, accurately modeling transparency effects *such as refraction and reflection*.
- **Order-independent transparency:** Techniques that ensure correct rendering of transparent objects regardless of the order in which they are drawn, preventing issues such as sorting artifacts.

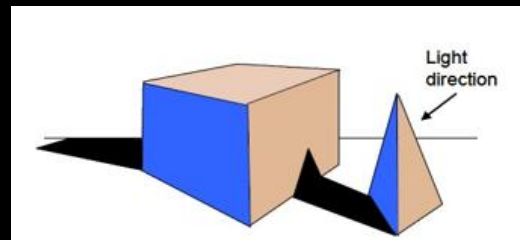
Shadow

Shadows are cast *when objects block light from reaching certain areas, creating areas of darkness on surfaces behind them*. In computer graphics, shadows are essential for creating realistic scenes and are commonly implemented using techniques such as:

1. **Shadow mapping:** A technique where the scene is rendered from the perspective of the light source to create a depth map, which is then used to determine which areas are in shadow when rendering from the camera's perspective.
2. **Shadow volumes:** A technique where shadow volumes are extruded from shadow-casting objects to determine which areas are in shadow based on the viewer's perspective.
3. **Ray tracing:** Ray tracing algorithms can accurately simulate the propagation of light rays, including the creation of shadows through the intersection of rays with objects in the scene.

Shadow

- Hidden surface method with light source at the view position can be used.
- The shadow area for all light sources are determined, and these shadows could be treated as a surface pattern arrays.
- Shadow *can help to create realism*. Without it, a cup, e.g., on a table may look as if the cup is floating in the air above the table.
- By applying *hidden-surface methods* with pretending that the position of a light source is the viewing position, we can find which surface sections cannot be "seen" from the light source => shadow areas.
- We usually display shadow areas with ambient-light intensity only.



Unit 8

Section 4: Polygon Rendering Methods : Constant Intensity Shading, Gouraud Shading

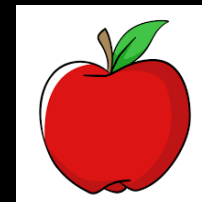


Polygon rendering methods (Surface Shading Method)

- **Polygon rendering** is a computer graphics technique that creates the illusion of a 3D object in a 2D space. It's used to create realistic images and animations in movies, video games, and simulations. Illumination model is applied to fill the interior of polygons.
- Shading is referred to as the implementation of the illumination model at the pixel points or polygon surfaces of the graphics objects. Shading model is used to compute the intensities and colors to display the surface. The shading model has two primary ingredients: properties of the surface and properties of the illumination falling on it
- Two ways of **polygon surface rendering**:
 - Single intensity for all points in a polygon
 - Interpolation of intensities for each point in a polygon

Methods:

1. Constant Intensity Shading
2. Gouraud Shading
3. Phong Shading



1. Constant Intensity Shading

- Fast and simple method for rendering an object with polygon surface
- Each polygon shaded with single intensity calculated for the polygon

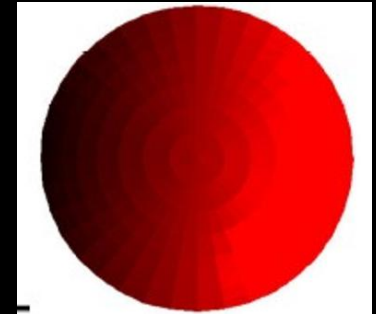
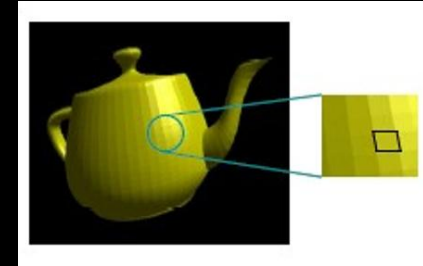
Procedure

1. Take a point on the object surface and calculate the intensity
2. Render the surface with same intensity throughout the surface
3. Repeat above procedure for each polygon surface

ASSUMPTIONS

1. Object is a polyhedron
2. light sources should be sufficiently (i.e. N.L and attenuation function are constant)
3. Viewing position is sufficiently far (i.e. V.R is constant over the surface)

DRAWBACK: intensity discontinuity at the edges of polygons



2. Gouraud Shading

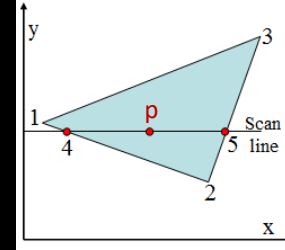
- Intensity interpolation method
- Renders a polygon surface by linearly interpolating intensity values across the surface.
- Intensity discontinuity at the edges of polygons is eliminated by matching intensity values of each polygon with adjacent polygons

Procedure

1. Determine the average unit normal vector at each polygon vertex
2. Calculate each of the vertex intensities by applying an illumination model
3. Linearly interpolate the vertex intensities over the polygon surface

2. Gouraud Shading

- For Gouraud shading, the intensity at point 4 is linearly interpolated from the intensities at vertices 1 and 2.
- The intensity at point 5 is linearly interpolated from intensities at vertices 2 and 3.
- An interior point p is then assigned an intensity value that is linearly interpolated from intensities at positions 4 and 5.

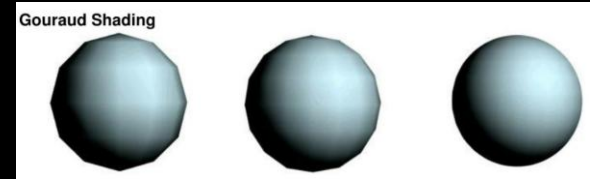


$$I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2 \quad I_p = \frac{x_5 - x_p}{x_5 - x_4} I_4 + \frac{x_p - x_4}{x_5 - x_4} I_5$$

2. Gouraud Shading

Advantages:

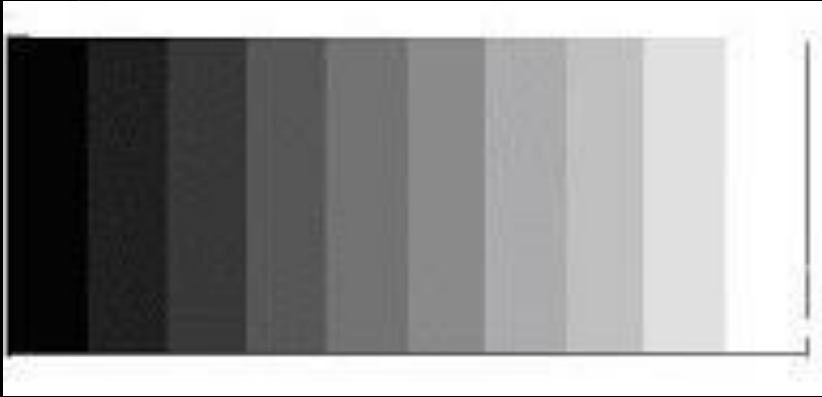
- **Eliminates Intensity Discontinuities:** Removes the blocky, stepped look of flat shading. It produces a smooth appearance across polygon boundaries by matching edge intensities.
- **Highly Efficient and optimized:** Computes lighting equations only at the vertices. It uses fast linear interpolation to fill internal pixels, making it computationally cheap. Well-supported by modern graphics hardware pipelines, rendering it ideal for real-time application performance.



Disadvantages

- **Mach Banding Effect:** Linear intensity interpolation causes sharp gradient changes at vertices, creating artificial bright or dark streaks on the surface.
- **Distorted Specular Highlights:** Specular highlights can appear with irregular, anomalous shapes. If a highlight falls in the middle of a large polygon without hitting a vertex, it can be missed entirely.

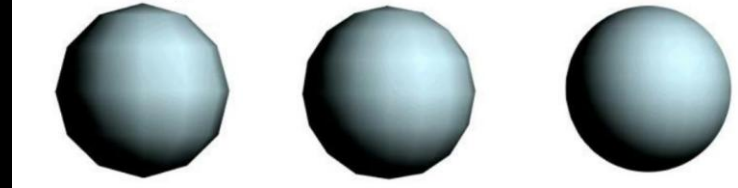
Mach Band



- **Mach Band effect** is a visual artifact where **artificial bright or dark streaks** appear along the edges of adjacent polygons.
- This happens because of **linear intensity interpolation** across polygon surfaces (as in Gouraud shading).



Gouraud Shading



Unit 8

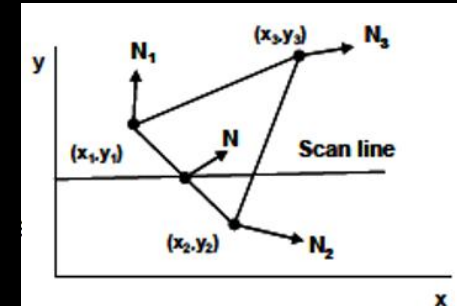
Section 5: Polygon Rendering Methods : Phong Shading, Gouraud Vs Phong Shading, Fast Phong

3. Phong Shading

- A more accurate method for rendering a polygon surface is Phong shading, or normal vector interpolation shading which first interpolate normal vectors and then apply the illumination model to each surface point. It displays more realistic highlights on a surface and greatly reduces the Mach-band effect.

A polygon surface is rendered using Phong shading by carrying out the following steps:

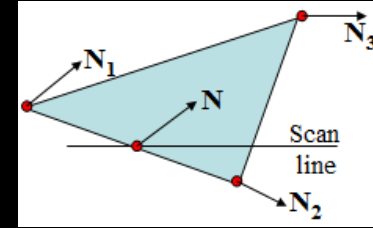
- Determine the average unit normal vector at each polygon vertex.
- Linearly interpolate the vertex normals over the surface of the polygon.
- Apply an illumination model along each scan line to calculate projected pixel intensities for the surface points.



3. Phong Shading

- The normal vector $\mathbf{N}$ for the scan-line intersection point along the edge between vertices 1 and 2 can be obtained by vertically interpolating between edge endpoint normal:

$$\mathbf{N} = \frac{y - y_2}{y_1 - y_2} \mathbf{N}_1 + \frac{y_1 - y}{y_1 - y_2} \mathbf{N}_2$$



- Incremental methods are used to evaluate normal between scan lines and along each individual scan line (as in Gouraud). At each pixel position along a scan line the illumination model is applied to determine the surface intensity at that point. Intensity calculations using an **approximated normal vector** at each point along the scan line produce more accurate results than the **direct interpolation of intensities**, as in Gouraud shading but it requires considerably more calculations.

3. Phong Shading

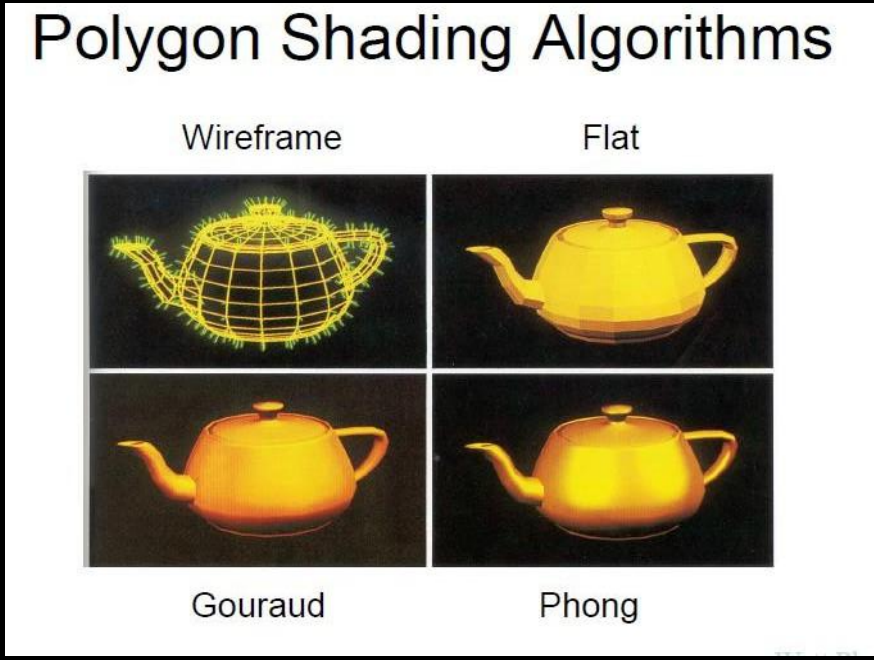
Advantages:

1. Some dark and bright intensity streaks known as Mach bands appear on the surface due to the linear intensity interpolation, this method reduces the Mach band effects and displays more realistic highlights.
2. It is more accurate as compared to the Gouraud Shading.

Disadvantages:

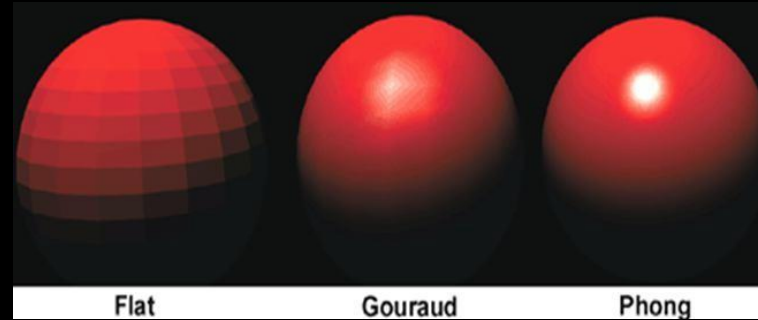
1. It is slower as compared to the Gouraud Shading.
2. It requires more calculation; hence it increases the cost of shading at each successive step.

Polygon Shading Algorithms



Gouraud Vs Phong Shading

- Gouraud shading is faster than Phong shading
- Phong shading is more accurate



Gouraud Vs Phong Shading

Gouraud Shading	Phong Shading
Gouraud shading was introduced by Henri Gouraud in 1970.	Phong Shading was introduced by Phong Bui Tuong in 1973.
Evaluate the illumination at border vertices and interpolates.	Evaluate the illumination at every point of the polygon surface.
Interpolates colors along edges and scanline.	Interpolates normals instead of colors.
Cheaper than the Phong Shading.	More costly than Gouraud Shading.
Lighting Equation used at each Vertex.	Lighting Equation used at each Pixel.
Requires average processing and less time.	Requires high processing and more time.
Gouraud Shading produces average quality.	Phong Shading produces good quality.
It gives less accurate results.	It gives more accurate results.
Introduces Mach band effect	Reduces Mach band effect

Fast Phong Shading

Fast Phong shading aims to achieve similar results to Phong shading but with fewer computations, *making it suitable for real-time rendering in games and other interactive applications*. Here's how it achieves speed improvements:

- **Approximations:** It might use approximations for some calculations, like replacing square roots with pre-calculated lookup tables or simplifying lighting models.
- **Reduced Math:** Techniques like pre-calculating lighting terms for specific materials or light sources can be employed to minimize calculations per pixel.

Key Differences:

- **Computational Cost:** Fast Phong shading is faster due to its optimizations.
- **Accuracy:** Regular Phong shading might offer slightly more accurate results due to its more complex calculations.

Fast Phong Shading

Pros of Fast Phong Shading:

- **Faster Rendering:** Ideal for real-time applications where smooth shading is desired without sacrificing performance.
- **Suitable for Hardware Acceleration:** Modern graphics cards (GPUs) are optimized for handling these types of calculations efficiently.

Cons of Fast Phong Shading:

- **Potential Loss of Accuracy:** The approximations used can introduce slight visual artifacts compared to full Phong shading.
- **May Require More Tuning:** Finding the right balance between speed and accuracy might involve adjustments to lighting parameters or material properties.

Questions

1. Define intensity attenuation. Distinguish between Gouraud shading and Phong shading model.
2. What is constant intensity shading? Compare Phong shading and fast Phong shading.
3. Explain Phong Shading Method with its advantage and disadvantage.
4. Write short notes on: Intensity Attenuation, Transparency, Color consideration and shadow.
5. Define illumination model. Compare between diffuse reflection and specular reflection.
6. What is specular reflection model of Phong? Explain shadows and intensity attenuation in illumination modelling?

THANK YOU
Any Queries ?